# PG No. 13 $C_{4v}$ $4mm$ [ tetragonal ] (internal axial octupole)

* Harmonics for rank 0

$\vec{\mathbb{G}}_0^{(3,3)}[g](A_2)$

** symmetry

1

** expression

$$\frac{\sqrt{105}G_{xyz}xyz}{7}+\frac{\sqrt{7}G_x^\alpha x\left(2x^2-3y^2-3z^2\right)}{14}+\frac{\sqrt{105}G_x^\beta x\left(y-z\right)\left(y+z\right)}{14}-\frac{\sqrt{7}G_y^\alpha y\left(3x^2-2y^2+3z^2\right)}{14}$$
$$-\frac{\sqrt{105}G_y^\beta y\left(x-z\right)\left(x+z\right)}{14}-\frac{\sqrt{7}G_z^\alpha z\left(3x^2+3y^2-2z^2\right)}{14}+\frac{\sqrt{105}G_z^\beta z\left(x-y\right)\left(x+y\right)}{14}$$

* Harmonics for rank 1

$\vec{\mathbb{Q}}_1^{(3,2)}[g](A_1)$

** symmetry

$z$

** expression

$$-\frac{\sqrt{105}G_{xyz}z\left(x-y\right)\left(x+y\right)}{14}+\frac{3\sqrt{7}G_x^\alpha y\left(4x^2-y^2-z^2\right)}{28}-\frac{\sqrt{105}G_x^\beta y\left(2x^2-y^2+z^2\right)}{28}$$
$$+\frac{3\sqrt{7}G_y^\alpha x\left(x^2-4y^2+z^2\right)}{28}+\frac{\sqrt{105}G_y^\beta x\left(x^2-2y^2-z^2\right)}{28}+\frac{\sqrt{105}G_z^\beta xyz}{7}$$

$\vec{\mathbb{G}}_1^{(3,1)}[g](A_2)$

** symmetry

$z$

** expression

$$\frac{\sqrt{21}G_{xyz}xy}{7}-\frac{3\sqrt{35}G_x^\alpha xz}{35}-\frac{\sqrt{21}G_x^\beta xz}{7}-\frac{3\sqrt{35}G_y^\alpha yz}{35}+\frac{\sqrt{21}G_y^\beta yz}{7}-\frac{3\sqrt{35}G_z^\alpha\left(x^2+y^2-2z^2\right)}{70}+\frac{\sqrt{21}G_z^\beta\left(x-y\right)\left(x+y\right)}{14}$$

$\vec{\mathbb{G}}_1^{(3,3)}[g](A_2)$

** symmetry

$z$

** expression

$$-\frac{\sqrt{35}G_{xyz}xy\left(x^2+y^2-6z^2\right)}{14}+\frac{5\sqrt{21}G_x^\alpha xz\left(4x^2-3y^2-3z^2\right)}{84}+\frac{\sqrt{35}G_x^\beta xz\left(2x^2+9y^2-5z^2\right)}{28}-\frac{5\sqrt{21}G_y^\alpha yz\left(3x^2-4y^2+3z^2\right)}{84}$$
$$-\frac{\sqrt{35}G_y^\beta yz\left(9x^2+2y^2-5z^2\right)}{28}+\frac{\sqrt{21}G_z^\alpha\left(3x^4+6x^2y^2-24x^2z^2+3y^4-24y^2z^2+8z^4\right)}{84}-\frac{\sqrt{35}G_z^\beta\left(x-y\right)\left(x+y\right)\left(x^2+y^2-6z^2\right)}{28}$$

$\vec{\mathbb{Q}}_{1,1}^{(3,2)}[g](E)$, $\vec{\mathbb{Q}}_{1,2}^{(3,2)}[g](E)$

** symmetry

$x$

$y$

** expression

$$-\frac{\sqrt{105}G_{xyz}x\left(y-z\right)\left(y+z\right)}{14}+\frac{\sqrt{105}G_x^\beta xyz}{7}-\frac{3\sqrt{7}G_y^\alpha z\left(x^2-4y^2+z^2\right)}{28}$$
$$-\frac{\sqrt{105}G_y^\beta z\left(x^2+2y^2-z^2\right)}{28}+\frac{3\sqrt{7}G_z^\alpha y\left(x^2+y^2-4z^2\right)}{28}-\frac{\sqrt{105}G_z^\beta y\left(x^2-y^2+2z^2\right)}{28}$$

$$\frac{\sqrt{105}G_{xyz}y\left(x-z\right)\left(x+z\right)}{14}-\frac{3\sqrt{7}G_x^\alpha z\left(4x^2-y^2-z^2\right)}{28}-\frac{\sqrt{105}G_x^\beta z\left(2x^2+y^2-z^2\right)}{28}$$
$$+\frac{\sqrt{105}G_y^\beta xyz}{7}-\frac{3\sqrt{7}G_z^\alpha x\left(x^2+y^2-4z^2\right)}{28}+\frac{\sqrt{105}G_z^\beta x\left(x^2-y^2-2z^2\right)}{28}$$

$\vec{\mathbb{G}}_{1,1}^{(3,1)}[g](E)$, $\vec{\mathbb{G}}_{1,2}^{(3,1)}[g](E)$

** symmetry

$-y$

$x$

** expression

$$-\frac{\sqrt{21}G_{xyz}xz}{7}+\frac{3\sqrt{35}G_x^\alpha xy}{35}-\frac{\sqrt{21}G_x^\beta xy}{7}+\frac{3\sqrt{35}G_y^\alpha\left(x^2-2y^2+z^2\right)}{70}+\frac{\sqrt{21}G_y^\beta\left(x-z\right)\left(x+z\right)}{14}+\frac{3\sqrt{35}G_z^\alpha yz}{35}+\frac{\sqrt{21}G_z^\beta yz}{7}$$

$$\frac{\sqrt{21}G_{xyz}yz}{7}+\frac{3\sqrt{35}G_x^{\alpha}\left(2x^2-y^2-z^2\right)}{70}+\frac{\sqrt{21}G_x^{\beta}\left(y-z\right)\left(y+z\right)}{14}-\frac{3\sqrt{35}G_y^{\alpha}xy}{35}-\frac{\sqrt{21}G_y^{\beta}xy}{7}-\frac{3\sqrt{35}G_z^{\alpha}xz}{35}+\frac{\sqrt{21}G_z^{\beta}xz}{7}$$

$\vec{\mathbb{G}}_{1,1}^{(3,3)}[g](E)$, $\vec{\mathbb{G}}_{1,2}^{(3,3)}[g](E)$

** symmetry

$$-y$$

$$x$$

** expression

$$\frac{\sqrt{35}G_{xyz}xz\left(x^2-6y^2+z^2\right)}{14}-\frac{5\sqrt{21}G_x^{\alpha}xy\left(4x^2-3y^2-3z^2\right)}{84}+\frac{\sqrt{35}G_x^{\beta}xy\left(2x^2-5y^2+9z^2\right)}{28}-\frac{\sqrt{21}G_y^{\alpha}\left(3x^4-24x^2y^2+6x^2z^2+8y^4-24y^2z^2+3z^4\right)}{84}$$
$$-\frac{\sqrt{35}G_y^{\beta}\left(x-z\right)\left(x+z\right)\left(x^2-6y^2+z^2\right)}{28}+\frac{5\sqrt{21}G_z^{\alpha}yz\left(3x^2+3y^2-4z^2\right)}{84}-\frac{\sqrt{35}G_z^{\beta}yz\left(9x^2-5y^2+2z^2\right)}{28}$$

$$\frac{\sqrt{35}G_{xyz}yz\left(6x^2-y^2-z^2\right)}{14}+\frac{\sqrt{21}G_x^{\alpha}\left(8x^4-24x^2y^2-24x^2z^2+3y^4+6y^2z^2+3z^4\right)}{84}+\frac{\sqrt{35}G_x^{\beta}\left(y-z\right)\left(y+z\right)\left(6x^2-y^2-z^2\right)}{28}$$
$$-\frac{5\sqrt{21}G_y^{\alpha}xy\left(3x^2-4y^2+3z^2\right)}{84}-\frac{\sqrt{35}G_y^{\beta}xy\left(5x^2-2y^2-9z^2\right)}{28}-\frac{5\sqrt{21}G_z^{\alpha}xz\left(3x^2+3y^2-4z^2\right)}{84}+\frac{\sqrt{35}G_z^{\beta}xz\left(5x^2-9y^2-2z^2\right)}{28}$$

* Harmonics for rank 2

$\vec{\mathbb{Q}}_2^{(3,0)}[g](A_1)$

** symmetry

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

** expression

$$-\frac{\sqrt{210}G_{xyz}\left(x-y\right)\left(x+y\right)}{28}-\frac{3\sqrt{14}G_x^{\alpha}yz}{28}-\frac{\sqrt{210}G_x^{\beta}yz}{28}+\frac{3\sqrt{14}G_y^{\alpha}xz}{28}-\frac{\sqrt{210}G_y^{\beta}xz}{28}+\frac{\sqrt{210}G_z^{\beta}xy}{14}$$

$\vec{\mathbb{Q}}_2^{(3,2)}[g](A_1)$

** symmetry

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

** expression

$$\frac{\sqrt{105}G_{xyz}\left(x-y\right)\left(x+y\right)\left(x^2+y^2-6z^2\right)}{42}+\frac{5\sqrt{7}G_x^{\alpha}yz\left(6x^2-y^2-z^2\right)}{28}-\frac{\sqrt{105}G_x^{\beta}yz\left(12x^2-9y^2+5z^2\right)}{84}$$
$$+\frac{5\sqrt{7}G_y^{\alpha}xz\left(x^2-6y^2+z^2\right)}{28}+\frac{\sqrt{105}G_y^{\beta}xz\left(9x^2-12y^2-5z^2\right)}{84}-\frac{\sqrt{105}G_z^{\beta}xy\left(x^2+y^2-6z^2\right)}{21}$$

$\vec{\mathbb{G}}_2^{(3,-1)}[g](A_2)$

** symmetry

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

** expression

$$-\frac{\sqrt{21}G_x^{\alpha}x}{14}-\frac{\sqrt{35}G_x^{\beta}x}{14}-\frac{\sqrt{21}G_y^{\alpha}y}{14}+\frac{\sqrt{35}G_y^{\beta}y}{14}+\frac{\sqrt{21}G_z^{\alpha}z}{7}$$

$\vec{\mathbb{G}}_2^{(3,1)}[g](A_2)$

** symmetry

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

** expression

$$-\frac{\sqrt{21}G_x^{\alpha}x\left(4x^2-21y^2+9z^2\right)}{84}+\frac{\sqrt{35}G_x^{\beta}x\left(2x^2-3y^2-3z^2\right)}{28}+\frac{\sqrt{21}G_y^{\alpha}y\left(21x^2-4y^2-9z^2\right)}{84}$$
$$+\frac{\sqrt{35}G_y^{\beta}y\left(3x^2-2y^2+3z^2\right)}{28}-\frac{\sqrt{21}G_z^{\alpha}z\left(3x^2+3y^2-2z^2\right)}{21}$$

$\vec{\mathbb{G}}_2^{(3,3)}[g](A_2)$

** symmetry

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

** expression

$$-\frac{3\sqrt{770}G_{xyz}xyz\left(x^2+y^2-2z^2\right)}{44}-\frac{5\sqrt{462}G_x^{\alpha}x\left(4x^4+x^2y^2-41x^2z^2-3y^4+15y^2z^2+18z^4\right)}{1848}$$
$$-\frac{\sqrt{770}G_x^{\beta}x\left(2x^4+11x^2y^2-31x^2z^2+9y^4-87y^2z^2+30z^4\right)}{616}$$
$$+\frac{5\sqrt{462}G_y^{\alpha}y\left(3x^4-x^2y^2-15x^2z^2-4y^4+41y^2z^2-18z^4\right)}{1848}+\frac{\sqrt{770}G_y^{\beta}y\left(9x^4+11x^2y^2-87x^2z^2+2y^4-31y^2z^2+30z^4\right)}{616}$$
$$+\frac{5\sqrt{462}G_z^{\alpha}z\left(15x^4+30x^2y^2-40x^2z^2+15y^4-40y^2z^2+8z^4\right)}{1848}-\frac{3\sqrt{770}G_z^{\beta}z\left(x-y\right)\left(x+y\right)\left(x^2+y^2-2z^2\right)}{88}$$

$\vec{\mathbb{Q}}_2^{(3,0)}[g](B_1)$
** symmetry

$$\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$$

** expression

$$-\frac{\sqrt{70}G_{xyz}\left(x^2+y^2-2z^2\right)}{28}-\frac{\sqrt{42}G_x^{\alpha}yz}{28}+\frac{3\sqrt{70}G_x^{\beta}yz}{28}-\frac{\sqrt{42}G_y^{\alpha}xz}{28}-\frac{3\sqrt{70}G_y^{\beta}xz}{28}+\frac{\sqrt{42}G_z^{\alpha}xy}{14}$$

$\vec{\mathbb{Q}}_2^{(3,2)}[g](B_1)$
** symmetry

$$\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$$

** expression

$$\frac{\sqrt{35}G_{xyz}\left(x^4-12x^2y^2+6x^2z^2+y^4+6y^2z^2-2z^4\right)}{42}+\frac{5\sqrt{21}G_x^{\alpha}yz\left(6x^2-y^2-z^2\right)}{84}+\frac{\sqrt{35}G_x^{\beta}yz\left(36x^2+y^2-13z^2\right)}{84}$$
$$-\frac{5\sqrt{21}G_y^{\alpha}xz\left(x^2-6y^2+z^2\right)}{84}-\frac{\sqrt{35}G_y^{\beta}xz\left(x^2+36y^2-13z^2\right)}{84}+\frac{5\sqrt{21}G_z^{\alpha}xy\left(x^2+y^2-6z^2\right)}{42}-\frac{\sqrt{35}G_z^{\beta}xy\left(x-y\right)\left(x+y\right)}{6}$$

$\vec{\mathbb{G}}_2^{(3,-1)}[g](B_1)$
** symmetry

$$\sqrt{3}xy$$

** expression

$$\frac{\sqrt{105}G_{xyz}z}{21}-\frac{\sqrt{7}G_x^{\alpha}y}{7}+\frac{\sqrt{105}G_x^{\beta}y}{21}-\frac{\sqrt{7}G_y^{\alpha}x}{7}-\frac{\sqrt{105}G_y^{\beta}x}{21}$$

$\vec{\mathbb{G}}_2^{(3,1)}[g](B_1)$
** symmetry

$$\sqrt{3}xy$$

** expression

$$\frac{\sqrt{105}G_{xyz}z\left(3x^2+3y^2-2z^2\right)}{42}+\frac{\sqrt{7}G_x^{\alpha}y\left(4x^2-y^2-z^2\right)}{28}+\frac{\sqrt{105}G_x^{\beta}y\left(6x^2+y^2-9z^2\right)}{84}$$
$$-\frac{\sqrt{7}G_y^{\alpha}x\left(x^2-4y^2+z^2\right)}{28}-\frac{\sqrt{105}G_y^{\beta}x\left(x^2+6y^2-9z^2\right)}{84}-\frac{5\sqrt{7}G_z^{\alpha}xyz}{7}$$

$\vec{\mathbb{G}}_2^{(3,3)}[g](B_1)$
** symmetry

$$\sqrt{3}xy$$

** expression

$$-\frac{\sqrt{2310}G_{xyz}z\left(6x^4-51x^2y^2+5x^2z^2+6y^4+5y^2z^2-z^4\right)}{462}+\frac{5\sqrt{154}G_x^{\alpha}y\left(8x^4-12x^2y^2-12x^2z^2+y^4+2y^2z^2+z^4\right)}{308}$$
$$-\frac{\sqrt{2310}G_x^{\beta}y\left(12x^4-46x^2y^2+66x^2z^2+5y^4-4y^2z^2-9z^4\right)}{924}+\frac{5\sqrt{154}G_y^{\alpha}x\left(x^4-12x^2y^2+2x^2z^2+8y^4-12y^2z^2+z^4\right)}{308}$$
$$+\frac{\sqrt{2310}G_y^{\beta}x\left(5x^4-46x^2y^2-4x^2z^2+12y^4+66y^2z^2-9z^4\right)}{924}-\frac{5\sqrt{154}G_z^{\alpha}xyz\left(x^2+y^2-2z^2\right)}{44}+\frac{3\sqrt{2310}G_z^{\beta}xyz\left(x-y\right)\left(x+y\right)}{44}$$

$\vec{\mathbb{Q}}_2^{(3,0)}[g](B_2)$
** symmetry

$$\sqrt{3}xy$$

** expression

$$-\frac{\sqrt{42}G_x^\alpha xz}{7}+\frac{\sqrt{42}G_y^\alpha yz}{7}-\frac{\sqrt{42}G_z^\alpha (x-y)(x+y)}{28}+\frac{\sqrt{70}G_z^\beta \left(x^2+y^2-2z^2\right)}{28}$$

$\vec{\mathbb{Q}}_2^{(3,2)}[g](B_2)$
** symmetry

$$\sqrt{3}xy$$

** expression

$$\frac{\sqrt{35}G_{xyz}xy(x-y)(x+y)}{6}-\frac{5\sqrt{21}G_x^\alpha xz\left(4x^2-3y^2-3z^2\right)}{84}-\frac{\sqrt{35}G_x^\beta xz\left(2x^2-3y^2-z^2\right)}{12}-\frac{5\sqrt{21}G_y^\alpha yz\left(3x^2-4y^2+3z^2\right)}{84}$$
$$+\frac{\sqrt{35}G_y^\beta yz\left(3x^2-2y^2+z^2\right)}{12}-\frac{5\sqrt{21}G_z^\alpha (x-y)(x+y)\left(x^2+y^2-6z^2\right)}{84}+\frac{\sqrt{35}G_z^\beta \left(5x^4-18x^2y^2-12x^2z^2+5y^4-12y^2z^2+4z^4\right)}{84}$$

$\vec{\mathbb{G}}_2^{(3,-1)}[g](B_2)$
** symmetry

$$\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$$

** expression

$$\frac{3\sqrt{7}G_x^\alpha x}{14}-\frac{\sqrt{105}G_x^\beta x}{42}-\frac{3\sqrt{7}G_y^\alpha y}{14}-\frac{\sqrt{105}G_y^\beta y}{42}+\frac{\sqrt{105}G_z^\beta z}{21}$$

$\vec{\mathbb{G}}_2^{(3,1)}[g](B_2)$
** symmetry

$$\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$$

** expression

$$\frac{\sqrt{7}G_x^\alpha x\left(4x^2-y^2-11z^2\right)}{28}+\frac{\sqrt{105}G_x^\beta x\left(2x^2-3y^2-3z^2\right)}{84}+\frac{\sqrt{7}G_y^\alpha y\left(x^2-4y^2+11z^2\right)}{28}$$
$$-\frac{\sqrt{105}G_y^\beta y\left(3x^2-2y^2+3z^2\right)}{84}-\frac{5\sqrt{7}G_z^\alpha z(x-y)(x+y)}{14}+\frac{\sqrt{105}G_z^\beta z\left(3x^2+3y^2-2z^2\right)}{42}$$

$\vec{\mathbb{G}}_2^{(3,3)}[g](B_2)$
** symmetry

$$\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$$

** expression

$$\frac{3\sqrt{2310}G_{xyz}xyz(x-y)(x+y)}{44}+\frac{5\sqrt{154}G_x^\alpha x\left(4x^4-27x^2y^2-13x^2z^2+11y^4+15y^2z^2+4z^4\right)}{616}$$
$$-\frac{\sqrt{2310}G_x^\beta x\left(2x^4-73x^2y^2+53x^2z^2+51y^4-87y^2z^2-12z^4\right)}{1848}-\frac{5\sqrt{154}G_y^\alpha y\left(11x^4-27x^2y^2+15x^2z^2+4y^4-13y^2z^2+4z^4\right)}{616}$$
$$-\frac{\sqrt{2310}G_y^\beta y\left(51x^4-73x^2y^2-87x^2z^2+2y^4+53y^2z^2-12z^4\right)}{1848}-\frac{5\sqrt{154}G_z^\alpha z(x-y)(x+y)\left(x^2+y^2-2z^2\right)}{88}$$
$$+\frac{\sqrt{2310}G_z^\beta z\left(39x^4-174x^2y^2-20x^2z^2+39y^4-20y^2z^2+4z^4\right)}{1848}$$

$\vec{\mathbb{Q}}_{2,1}^{(3,0)}[g](E)$, $\vec{\mathbb{Q}}_{2,2}^{(3,0)}[g](E)$
** symmetry

$$\sqrt{3}xz$$

$$\sqrt{3}yz$$

** expression

$$\frac{\sqrt{42}G_x^\alpha xy}{7}+\frac{\sqrt{42}G_y^\alpha (x-z)(x+z)}{28}+\frac{\sqrt{70}G_y^\beta \left(x^2-2y^2+z^2\right)}{28}-\frac{\sqrt{42}G_z^\alpha yz}{7}$$

$$-\frac{\sqrt{42}G_x^\alpha (y-z)(y+z)}{28}-\frac{\sqrt{70}G_x^\beta \left(2x^2-y^2-z^2\right)}{28}-\frac{\sqrt{42}G_y^\alpha xy}{7}+\frac{\sqrt{42}G_z^\alpha xz}{7}$$

$\vec{\mathbb{Q}}_{2,1}^{(3,2)}[g](E)$, $\vec{\mathbb{Q}}_{2,2}^{(3,2)}[g](E)$
** symmetry

$$\sqrt{3}xz$$

$\sqrt{3}yz$

** expression

$$-\frac{\sqrt{35}G_{xyz}xz\left(x-z\right)\left(x+z\right)}{6}+\frac{5\sqrt{21}G_{x}^{\alpha}xy\left(4x^{2}-3y^{2}-3z^{2}\right)}{84}-\frac{\sqrt{35}G_{x}^{\beta}xy\left(2x^{2}-y^{2}-3z^{2}\right)}{12}+\frac{5\sqrt{21}G_{y}^{\alpha}\left(x-z\right)\left(x+z\right)\left(x^{2}-6y^{2}+z^{2}\right)}{84}$$
$$+\frac{\sqrt{35}G_{y}^{\beta}\left(5x^{4}-12x^{2}y^{2}-18x^{2}z^{2}+4y^{4}-12y^{2}z^{2}+5z^{4}\right)}{84}+\frac{5\sqrt{21}G_{z}^{\alpha}yz\left(3x^{2}+3y^{2}-4z^{2}\right)}{84}+\frac{\sqrt{35}G_{z}^{\beta}yz\left(3x^{2}+y^{2}-2z^{2}\right)}{12}$$

$$\frac{\sqrt{35}G_{xyz}yz\left(y-z\right)\left(y+z\right)}{6}+\frac{5\sqrt{21}G_{x}^{\alpha}\left(y-z\right)\left(y+z\right)\left(6x^{2}-y^{2}-z^{2}\right)}{84}+\frac{\sqrt{35}G_{x}^{\beta}\left(4x^{4}-12x^{2}y^{2}-12x^{2}z^{2}+5y^{4}-18y^{2}z^{2}+5z^{4}\right)}{84}$$
$$+\frac{5\sqrt{21}G_{y}^{\alpha}xy\left(3x^{2}-4y^{2}+3z^{2}\right)}{84}+\frac{\sqrt{35}G_{y}^{\beta}xy\left(x^{2}-2y^{2}+3z^{2}\right)}{12}-\frac{5\sqrt{21}G_{z}^{\alpha}xz\left(3x^{2}+3y^{2}-4z^{2}\right)}{84}+\frac{\sqrt{35}G_{z}^{\beta}xz\left(x^{2}+3y^{2}-2z^{2}\right)}{12}$$

$\vec{\mathbb{G}}_{2,1}^{(3,-1)}[g](E)$, $\vec{\mathbb{G}}_{2,2}^{(3,-1)}[g](E)$
** symmetry

$\sqrt{3}yz$

$-\sqrt{3}xz$

** expression

$$\frac{\sqrt{105}G_{xyz}x}{21}-\frac{\sqrt{7}G_{y}^{\alpha}z}{7}+\frac{\sqrt{105}G_{y}^{\beta}z}{21}-\frac{\sqrt{7}G_{z}^{\alpha}y}{7}-\frac{\sqrt{105}G_{z}^{\beta}y}{21}$$

$$-\frac{\sqrt{105}G_{xyz}y}{21}+\frac{\sqrt{7}G_{x}^{\alpha}z}{7}+\frac{\sqrt{105}G_{x}^{\beta}z}{21}+\frac{\sqrt{7}G_{z}^{\alpha}x}{7}-\frac{\sqrt{105}G_{z}^{\beta}x}{21}$$

$\vec{\mathbb{G}}_{2,1}^{(3,1)}[g](E)$, $\vec{\mathbb{G}}_{2,2}^{(3,1)}[g](E)$
** symmetry

$\sqrt{3}yz$

$-\sqrt{3}xz$

** expression

$$-\frac{\sqrt{105}G_{xyz}x\left(2x^{2}-3y^{2}-3z^{2}\right)}{42}-\frac{5\sqrt{7}G_{x}^{\alpha}xyz}{7}-\frac{\sqrt{7}G_{y}^{\alpha}z\left(x^{2}-4y^{2}+z^{2}\right)}{28}$$
$$-\frac{\sqrt{105}G_{y}^{\beta}z\left(9x^{2}-6y^{2}-z^{2}\right)}{84}-\frac{\sqrt{7}G_{z}^{\alpha}y\left(x^{2}+y^{2}-4z^{2}\right)}{28}+\frac{\sqrt{105}G_{z}^{\beta}y\left(9x^{2}-y^{2}-6z^{2}\right)}{84}$$

$$-\frac{\sqrt{105}G_{xyz}y\left(3x^{2}-2y^{2}+3z^{2}\right)}{42}-\frac{\sqrt{7}G_{x}^{\alpha}z\left(4x^{2}-y^{2}-z^{2}\right)}{28}+\frac{\sqrt{105}G_{x}^{\beta}z\left(6x^{2}-9y^{2}+z^{2}\right)}{84}$$
$$+\frac{5\sqrt{7}G_{y}^{\alpha}xyz}{7}+\frac{\sqrt{7}G_{z}^{\alpha}x\left(x^{2}+y^{2}-4z^{2}\right)}{28}-\frac{\sqrt{105}G_{z}^{\beta}x\left(x^{2}-9y^{2}+6z^{2}\right)}{84}$$

$\vec{\mathbb{G}}_{2,1}^{(3,3)}[g](E)$, $\vec{\mathbb{G}}_{2,2}^{(3,3)}[g](E)$
** symmetry

$\sqrt{3}yz$

$-\sqrt{3}xz$

** expression

$$\frac{\sqrt{2310}G_{xyz}x\left(x^{4}-5x^{2}y^{2}-5x^{2}z^{2}-6y^{4}+51y^{2}z^{2}-6z^{4}\right)}{462}+\frac{5\sqrt{154}G_{x}^{\alpha}xyz\left(2x^{2}-y^{2}-z^{2}\right)}{44}+\frac{3\sqrt{2310}G_{x}^{\beta}xyz\left(y-z\right)\left(y+z\right)}{44}$$
$$+\frac{5\sqrt{154}G_{y}^{\alpha}z\left(x^{4}-12x^{2}y^{2}+2x^{2}z^{2}+8y^{4}-12y^{2}z^{2}+z^{4}\right)}{308}+\frac{\sqrt{2310}G_{y}^{\beta}z\left(9x^{4}-66x^{2}y^{2}+4x^{2}z^{2}-12y^{4}+46y^{2}z^{2}-5z^{4}\right)}{924}$$
$$+\frac{5\sqrt{154}G_{z}^{\alpha}y\left(x^{4}+2x^{2}y^{2}-12x^{2}z^{2}+y^{4}-12y^{2}z^{2}+8z^{4}\right)}{308}-\frac{\sqrt{2310}G_{z}^{\beta}y\left(9x^{4}+4x^{2}y^{2}-66x^{2}z^{2}-5y^{4}+46y^{2}z^{2}-12z^{4}\right)}{924}$$

$$\frac{\sqrt{2310}G_{xyz}y\left(6x^{4}+5x^{2}y^{2}-51x^{2}z^{2}-y^{4}+5y^{2}z^{2}+6z^{4}\right)}{462}-\frac{5\sqrt{154}G_{x}^{\alpha}z\left(8x^{4}-12x^{2}y^{2}-12x^{2}z^{2}+y^{4}+2y^{2}z^{2}+z^{4}\right)}{308}$$
$$-\frac{\sqrt{2310}G_{x}^{\beta}z\left(12x^{4}+66x^{2}y^{2}-46x^{2}z^{2}-9y^{4}-4y^{2}z^{2}+5z^{4}\right)}{924}+\frac{5\sqrt{154}G_{y}^{\alpha}xyz\left(x^{2}-2y^{2}+z^{2}\right)}{44}+\frac{3\sqrt{2310}G_{y}^{\beta}xyz\left(x-z\right)\left(x+z\right)}{44}$$
$$-\frac{5\sqrt{154}G_{z}^{\alpha}x\left(x^{4}+2x^{2}y^{2}-12x^{2}z^{2}+y^{4}-12y^{2}z^{2}+8z^{4}\right)}{308}+\frac{\sqrt{2310}G_{z}^{\beta}x\left(5x^{4}-4x^{2}y^{2}-46x^{2}z^{2}-9y^{4}+66y^{2}z^{2}+12z^{4}\right)}{924}$$

* Harmonics for rank 3

$\vec{\mathbb{Q}}_3^{(3,-2)}[g](A_1)$

** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$-\frac{\sqrt{3}G_x^\alpha y}{4}-\frac{\sqrt{5}G_x^\beta y}{4}+\frac{\sqrt{3}G_y^\alpha x}{4}-\frac{\sqrt{5}G_y^\beta x}{4}$$

$\vec{\mathbb{Q}}_3^{(3,0)}[g](A_1)$

** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$-\frac{\sqrt{10}G_{xyz}z\,(x-y)\,(x+y)}{4}-\frac{\sqrt{6}G_x^\alpha y\left(3x^2-2y^2+3z^2\right)}{12}+\frac{\sqrt{10}G_x^\beta y\,(x-z)\,(x+z)}{4}$$
$$-\frac{\sqrt{6}G_y^\alpha x\left(2x^2-3y^2-3z^2\right)}{12}+\frac{\sqrt{10}G_y^\beta x\,(y-z)\,(y+z)}{4}+\frac{\sqrt{10}G_z^\beta xyz}{2}$$

$\vec{\mathbb{Q}}_3^{(3,2)}[g](A_1)$

** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$\frac{7\sqrt{55}G_{xyz}z\,(x-y)\,(x+y)\left(x^2+y^2-2z^2\right)}{44}-\frac{5\sqrt{33}G_x^\alpha y\left(6x^4+5x^2y^2-51x^2z^2-y^4+5y^2z^2+6z^4\right)}{264}$$
$$+\frac{\sqrt{55}G_x^\beta y\left(4x^4+x^2y^2-27x^2z^2-3y^4+29y^2z^2-10z^4\right)}{88}-\frac{5\sqrt{33}G_y^\alpha x\left(x^4-5x^2y^2-5x^2z^2-6y^4+51y^2z^2-6z^4\right)}{264}$$
$$-\frac{\sqrt{55}G_y^\beta x\left(3x^4-x^2y^2-29x^2z^2-4y^4+27y^2z^2+10z^4\right)}{88}-\frac{7\sqrt{55}G_z^\beta xyz\left(x^2+y^2-2z^2\right)}{22}$$

$\vec{\mathbb{G}}_3^{(3,-3)}[g](A_2)$

** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$G_z^\alpha$$

$\vec{\mathbb{G}}_3^{(3,-1)}[g](A_2)$

** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$-G_{xyz}xy-\frac{\sqrt{15}G_x^\alpha xz}{20}-\frac{G_x^\beta xz}{4}-\frac{\sqrt{15}G_y^\alpha yz}{20}+\frac{G_y^\beta yz}{4}-\frac{\sqrt{15}G_z^\alpha\left(x^2+y^2-2z^2\right)}{15}-\frac{G_z^\beta\,(x-y)\,(x+y)}{2}$$

$\vec{\mathbb{G}}_3^{(3,1)}[g](A_2)$

** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$\frac{\sqrt{330}G_{xyz}xy\left(x^2+y^2-6z^2\right)}{132}-\frac{5\sqrt{22}G_x^\alpha xz\left(x^2-6y^2+z^2\right)}{44}+\frac{\sqrt{330}G_x^\beta xz\left(9x^2-12y^2-5z^2\right)}{132}+\frac{5\sqrt{22}G_y^\alpha yz\left(6x^2-y^2-z^2\right)}{44}$$
$$+\frac{\sqrt{330}G_y^\beta yz\left(12x^2-9y^2+5z^2\right)}{132}+\frac{\sqrt{22}G_z^\alpha\left(3x^4+6x^2y^2-24x^2z^2+3y^4-24y^2z^2+8z^4\right)}{88}+\frac{\sqrt{330}G_z^\beta\,(x-y)\,(x+y)\left(x^2+y^2-6z^2\right)}{264}$$

$\vec{\mathbb{G}}_3^{(3,3)}[g](A_2)$

** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$\frac{7\sqrt{715}G_{xyz}xy\left(x^4+2x^2y^2-16x^2z^2+y^4-16y^2z^2+16z^4\right)}{572}-\frac{35\sqrt{429}G_x^{\alpha}xz\left(2x^4+x^2y^2-7x^2z^2-y^4+y^2z^2+2z^4\right)}{1144}$$
$$-\frac{7\sqrt{715}G_x^{\beta}xz\left(4x^4+17x^2y^2-19x^2z^2+13y^4-43y^2z^2+10z^4\right)}{1144}$$
$$+\frac{35\sqrt{429}G_y^{\alpha}yz\left(x^4-x^2y^2-x^2z^2-2y^4+7y^2z^2-2z^4\right)}{1144}+\frac{7\sqrt{715}G_y^{\beta}yz\left(13x^4+17x^2y^2-43x^2z^2+4y^4-19y^2z^2+10z^4\right)}{1144}$$
$$-\frac{5\sqrt{429}G_z^{\alpha}\left(5x^6+15x^4y^2-90x^4z^2+15x^2y^4-180x^2y^2z^2+120x^2z^4+5y^6-90y^4z^2+120y^2z^4-16z^6\right)}{3432}$$
$$+\frac{7\sqrt{715}G_z^{\beta}(x-y)(x+y)\left(x^4+2x^2y^2-16x^2z^2+y^4-16y^2z^2+16z^4\right)}{1144}$$

$\vec{\mathbb{Q}}_3^{(3,-2)}[g](B_1)$

** symmetry

$$\frac{\sqrt{15}x^2z}{2}-\frac{\sqrt{15}y^2z}{2}$$

** expression

$$\frac{\sqrt{3}G_{xyz}z}{3}+\frac{\sqrt{5}G_x^{\alpha}y}{4}+\frac{\sqrt{3}G_x^{\beta}y}{12}+\frac{\sqrt{5}G_y^{\alpha}x}{4}-\frac{\sqrt{3}G_y^{\beta}x}{12}$$

$\vec{\mathbb{Q}}_3^{(3,0)}[g](B_1)$

** symmetry

$$\frac{\sqrt{15}x^2z}{2}-\frac{\sqrt{15}y^2z}{2}$$

** expression

$$-\frac{\sqrt{6}G_{xyz}z\left(3x^2+3y^2-2z^2\right)}{12}+\frac{\sqrt{10}G_x^{\alpha}y(x-z)(x+z)}{4}+\frac{\sqrt{6}G_x^{\beta}y\left(3x^2-2y^2+3z^2\right)}{12}$$
$$+\frac{\sqrt{10}G_y^{\alpha}x(y-z)(y+z)}{4}+\frac{\sqrt{6}G_y^{\beta}x\left(2x^2-3y^2-3z^2\right)}{12}-\frac{\sqrt{10}G_z^{\alpha}xyz}{2}$$

$\vec{\mathbb{Q}}_3^{(3,2)}[g](B_1)$

** symmetry

$$\frac{\sqrt{15}x^2z}{2}-\frac{\sqrt{15}y^2z}{2}$$

** expression

$$-\frac{\sqrt{33}G_{xyz}z\left(15x^4+30x^2y^2-40x^2z^2+15y^4-40y^2z^2+8z^4\right)}{132}+\frac{\sqrt{55}G_x^{\alpha}y\left(10x^4-29x^2y^2+27x^2z^2+3y^4-y^2z^2-4z^4\right)}{88}$$
$$-\frac{\sqrt{33}G_x^{\beta}y\left(60x^4-55x^2y^2-195x^2z^2+11y^4-55y^2z^2+60z^4\right)}{264}+\frac{\sqrt{55}G_y^{\alpha}x\left(3x^4-29x^2y^2-x^2z^2+10y^4+27y^2z^2-4z^4\right)}{88}$$
$$+\frac{\sqrt{33}G_y^{\beta}x\left(11x^4-55x^2y^2-55x^2z^2+60y^4-195y^2z^2+60z^4\right)}{264}+\frac{7\sqrt{55}G_z^{\alpha}xyz\left(x^2+y^2-2z^2\right)}{22}$$

$\vec{\mathbb{G}}_3^{(3,-3)}[g](B_1)$

** symmetry

$$\sqrt{15}xyz$$

** expression

$$G_{xyz}$$

$\vec{\mathbb{G}}_3^{(3,-1)}[g](B_1)$

** symmetry

$$\sqrt{15}xyz$$

** expression

$$-G_x^{\alpha}yz-G_y^{\alpha}xz-G_z^{\alpha}xy$$

$\vec{\mathbb{G}}_3^{(3,1)}[g](B_1)$

** symmetry

$$\sqrt{15}xyz$$

** expression

$$-\frac{7\sqrt{22}G_{xyz}\left(x^4-3x^2y^2-3x^2z^2+y^4-3y^2z^2+z^4\right)}{66}-\frac{\sqrt{330}G_x^{\alpha}yz\left(6x^2-y^2-z^2\right)}{132}+\frac{35\sqrt{22}G_x^{\beta}yz(y-z)(y+z)}{132}$$
$$+\frac{\sqrt{330}G_y^{\alpha}xz\left(x^2-6y^2+z^2\right)}{132}-\frac{35\sqrt{22}G_y^{\beta}xz(x-z)(x+z)}{132}+\frac{\sqrt{330}G_z^{\alpha}xy\left(x^2+y^2-6z^2\right)}{132}+\frac{35\sqrt{22}G_z^{\beta}xy(x-y)(x+y)}{132}$$

$\vec{\mathbb{G}}_3^{(3,3)}[g](B_1)$

** symmetry

$$\sqrt{15}xyz$$

** expression

$$\frac{\sqrt{429}G_{xyz}\left(2x^6-15x^4y^2-15x^4z^2-15x^2y^4+180x^2y^2z^2-15x^2z^4+2y^6-15y^4z^2-15y^2z^4+2z^6\right)}{286}$$
$$+\frac{7\sqrt{715}G_x^{\alpha}yz\left(16x^4-16x^2y^2-16x^2z^2+y^4+2y^2z^2+z^4\right)}{572}+\frac{21\sqrt{429}G_x^{\beta}yz\,(y-z)\,(y+z)\left(10x^2-y^2-z^2\right)}{572}$$
$$+\frac{7\sqrt{715}G_y^{\alpha}xz\left(x^4-16x^2y^2+2x^2z^2+16y^4-16y^2z^2+z^4\right)}{572}+\frac{21\sqrt{429}G_y^{\beta}xz\,(x-z)\,(x+z)\left(x^2-10y^2+z^2\right)}{572}$$
$$+\frac{7\sqrt{715}G_z^{\alpha}xy\left(x^4+2x^2y^2-16x^2z^2+y^4-16y^2z^2+16z^4\right)}{572}-\frac{21\sqrt{429}G_z^{\beta}xy\,(x-y)\,(x+y)\left(x^2+y^2-10z^2\right)}{572}$$

$\vec{\mathbb{Q}}_3^{(3,-2)}[g](B_2)$

** symmetry

$$\sqrt{15}xyz$$

** expression

$$-\frac{\sqrt{3}G_x^{\beta}x}{3}-\frac{\sqrt{3}G_y^{\beta}y}{3}-\frac{\sqrt{3}G_z^{\beta}z}{3}$$

$\vec{\mathbb{Q}}_3^{(3,0)}[g](B_2)$

** symmetry

$$\sqrt{15}xyz$$

** expression

$$\frac{\sqrt{10}G_x^{\alpha}x\,(y-z)\,(y+z)}{4}-\frac{\sqrt{6}G_x^{\beta}x\left(2x^2-3y^2-3z^2\right)}{12}-\frac{\sqrt{10}G_y^{\alpha}y\,(x-z)\,(x+z)}{4}$$
$$+\frac{\sqrt{6}G_y^{\beta}y\left(3x^2-2y^2+3z^2\right)}{12}+\frac{\sqrt{10}G_z^{\alpha}z\,(x-y)\,(x+y)}{4}+\frac{\sqrt{6}G_z^{\beta}z\left(3x^2+3y^2-2z^2\right)}{12}$$

$\vec{\mathbb{Q}}_3^{(3,2)}[g](B_2)$

** symmetry

$$\sqrt{15}xyz$$

** expression

$$\frac{7\sqrt{55}G_x^{\alpha}x\,(y-z)\,(y+z)\left(2x^2-y^2-z^2\right)}{44}+\frac{\sqrt{33}G_x^{\beta}x\left(8x^4-40x^2y^2-40x^2z^2+15y^4+30y^2z^2+15z^4\right)}{132}$$
$$+\frac{7\sqrt{55}G_y^{\alpha}y\,(x-z)\,(x+z)\left(x^2-2y^2+z^2\right)}{44}+\frac{\sqrt{33}G_y^{\beta}y\left(15x^4-40x^2y^2+30x^2z^2+8y^4-40y^2z^2+15z^4\right)}{132}$$
$$-\frac{7\sqrt{55}G_z^{\alpha}z\,(x-y)\,(x+y)\left(x^2+y^2-2z^2\right)}{44}+\frac{\sqrt{33}G_z^{\beta}z\left(15x^4+30x^2y^2-40x^2z^2+15y^4-40y^2z^2+8z^4\right)}{132}$$

$\vec{\mathbb{G}}_3^{(3,-3)}[g](B_2)$

** symmetry

$$\frac{\sqrt{15}x^2z}{2}-\frac{\sqrt{15}y^2z}{2}$$

** expression

$$G_z^{\beta}$$

$\vec{\mathbb{G}}_3^{(3,-1)}[g](B_2)$

** symmetry

$$\frac{\sqrt{15}x^2z}{2}-\frac{\sqrt{15}y^2z}{2}$$

** expression

$$\frac{G_x^{\alpha}xz}{4}-\frac{\sqrt{15}G_x^{\beta}xz}{4}-\frac{G_y^{\alpha}yz}{4}-\frac{\sqrt{15}G_y^{\beta}yz}{4}-\frac{G_z^{\alpha}\,(x-y)\,(x+y)}{2}$$

$\vec{\mathbb{G}}_3^{(3,1)}[g](B_2)$

** symmetry

$$\frac{\sqrt{15}x^2z}{2}-\frac{\sqrt{15}y^2z}{2}$$

** expression

$$\frac{35\sqrt{22}G_{xyz}xy\left(x-y\right)\left(x+y\right)}{132}+\frac{\sqrt{330}G_{x}^{\alpha}xz\left(5x^{2}+12y^{2}-9z^{2}\right)}{132}-\frac{5\sqrt{22}G_{x}^{\beta}xz\left(x^{2}-6y^{2}+z^{2}\right)}{132}-\frac{\sqrt{330}G_{y}^{\alpha}yz\left(12x^{2}+5y^{2}-9z^{2}\right)}{132}$$
$$+\frac{5\sqrt{22}G_{y}^{\beta}yz\left(6x^{2}-y^{2}-z^{2}\right)}{132}+\frac{\sqrt{330}G_{z}^{\alpha}\left(x-y\right)\left(x+y\right)\left(x^{2}+y^{2}-6z^{2}\right)}{264}+\frac{7\sqrt{22}G_{z}^{\beta}\left(x^{4}-18x^{2}y^{2}+12x^{2}z^{2}+y^{4}+12y^{2}z^{2}-4z^{4}\right)}{264}$$

$\vec{\mathbb{G}}_{3}^{(3,3)}[g](B_2)$
** symmetry

$$\frac{\sqrt{15}x^{2}z}{2}-\frac{\sqrt{15}y^{2}z}{2}$$

** expression

$$-\frac{21\sqrt{429}G_{xyz}xy\left(x-y\right)\left(x+y\right)\left(x^{2}+y^{2}-10z^{2}\right)}{572}+\frac{7\sqrt{715}G_{x}^{\alpha}xz\left(10x^{4}-43x^{2}y^{2}-19x^{2}z^{2}+13y^{4}+17y^{2}z^{2}+4z^{4}\right)}{1144}$$
$$+\frac{7\sqrt{429}G_{x}^{\beta}xz\left(4x^{4}+35x^{2}y^{2}-25x^{2}z^{2}-35y^{4}+35y^{2}z^{2}+4z^{4}\right)}{1144}-\frac{7\sqrt{715}G_{y}^{\alpha}yz\left(13x^{4}-43x^{2}y^{2}+17x^{2}z^{2}+10y^{4}-19y^{2}z^{2}+4z^{4}\right)}{1144}$$
$$-\frac{7\sqrt{429}G_{y}^{\beta}yz\left(35x^{4}-35x^{2}y^{2}-35x^{2}z^{2}-4y^{4}+25y^{2}z^{2}-4z^{4}\right)}{1144}+\frac{7\sqrt{715}G_{z}^{\alpha}\left(x-y\right)\left(x+y\right)\left(x^{4}+2x^{2}y^{2}-16x^{2}z^{2}+y^{4}-16y^{2}z^{2}+16z^{4}\right)}{1144}$$
$$-\frac{\sqrt{429}G_{z}^{\beta}\left(13x^{6}-45x^{4}y^{2}-150x^{4}z^{2}-45x^{2}y^{4}+540x^{2}y^{2}z^{2}+60x^{2}z^{4}+13y^{6}-150y^{4}z^{2}+60y^{2}z^{4}-8z^{6}\right)}{1144}$$

$\vec{\mathbb{Q}}_{3,1}^{(3,-2)}[g](E,1)$, $\vec{\mathbb{Q}}_{3,2}^{(3,-2)}[g](E,1)$
** symmetry

$$x^{3}-\frac{3xy^{2}}{2}-\frac{3xz^{2}}{2}$$

$$-\frac{3x^{2}y}{2}+y^{3}-\frac{3yz^{2}}{2}$$

** expression

$$-\frac{\sqrt{3}G_{y}^{\alpha}z}{4}-\frac{\sqrt{5}G_{y}^{\beta}z}{4}+\frac{\sqrt{3}G_{z}^{\alpha}y}{4}-\frac{\sqrt{5}G_{z}^{\beta}y}{4}$$

$$\frac{\sqrt{3}G_{x}^{\alpha}z}{4}-\frac{\sqrt{5}G_{x}^{\beta}z}{4}-\frac{\sqrt{3}G_{z}^{\alpha}x}{4}-\frac{\sqrt{5}G_{z}^{\beta}x}{4}$$

$\vec{\mathbb{Q}}_{3,1}^{(3,-2)}[g](E,2)$, $\vec{\mathbb{Q}}_{3,2}^{(3,-2)}[g](E,2)$
** symmetry

$$\frac{\sqrt{15}xy^{2}}{2}-\frac{\sqrt{15}xz^{2}}{2}$$

$$\frac{\sqrt{15}x^{2}y}{2}-\frac{\sqrt{15}yz^{2}}{2}$$

** expression

$$\frac{\sqrt{3}G_{xyz}x}{3}+\frac{\sqrt{5}G_{y}^{\alpha}z}{4}+\frac{\sqrt{3}G_{y}^{\beta}z}{12}+\frac{\sqrt{5}G_{z}^{\alpha}y}{4}-\frac{\sqrt{3}G_{z}^{\beta}y}{12}$$

$$-\frac{\sqrt{3}G_{xyz}y}{3}-\frac{\sqrt{5}G_{x}^{\alpha}z}{4}+\frac{\sqrt{3}G_{x}^{\beta}z}{12}-\frac{\sqrt{5}G_{z}^{\alpha}x}{4}-\frac{\sqrt{3}G_{z}^{\beta}x}{12}$$

$\vec{\mathbb{Q}}_{3,1}^{(3,0)}[g](E,1)$, $\vec{\mathbb{Q}}_{3,2}^{(3,0)}[g](E,1)$
** symmetry

$$x^{3}-\frac{3xy^{2}}{2}-\frac{3xz^{2}}{2}$$

$$-\frac{3x^{2}y}{2}+y^{3}-\frac{3yz^{2}}{2}$$

** expression

$$-\frac{\sqrt{10}G_{xyz}x\left(y-z\right)\left(y+z\right)}{4}+\frac{\sqrt{10}G_{x}^{\beta}xyz}{2}-\frac{\sqrt{6}G_{y}^{\alpha}z\left(3x^{2}+3y^{2}-2z^{2}\right)}{12}$$
$$-\frac{\sqrt{10}G_{y}^{\beta}z\left(x-y\right)\left(x+y\right)}{4}+\frac{\sqrt{6}G_{z}^{\alpha}y\left(3x^{2}-2y^{2}+3z^{2}\right)}{12}-\frac{\sqrt{10}G_{z}^{\beta}y\left(x-z\right)\left(x+z\right)}{4}$$

$$\frac{\sqrt{10}G_{xyz}y\left(x-z\right)\left(x+z\right)}{4}+\frac{\sqrt{6}G_{x}^{\alpha}z\left(3x^{2}+3y^{2}-2z^{2}\right)}{12}+\frac{\sqrt{10}G_{x}^{\beta}z\left(x-y\right)\left(x+y\right)}{4}+\frac{\sqrt{10}G_{y}^{\beta}xyz}{2}+\frac{\sqrt{6}G_{z}^{\alpha}x\left(2x^{2}-3y^{2}-3z^{2}\right)}{12}-\frac{\sqrt{10}G_{z}^{\beta}x\left(y-z\right)\left(y+z\right)}{4}$$

$\vec{\mathbb{Q}}^{(3,0)}_{3,1}[g](E,2)$, $\vec{\mathbb{Q}}^{(3,0)}_{3,2}[g](E,2)$

** symmetry

$$\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$$

$$\frac{\sqrt{15}x^2y}{2} - \frac{\sqrt{15}yz^2}{2}$$

** expression

$$\frac{\sqrt{6}G_{xyz}x\left(2x^2-3y^2-3z^2\right)}{12} - \frac{\sqrt{10}G^{\alpha}_{x}xyz}{2} - \frac{\sqrt{10}G^{\alpha}_{y}z\left(x-y\right)\left(x+y\right)}{4} + \frac{\sqrt{6}G^{\beta}_{y}z\left(3x^2+3y^2-2z^2\right)}{12} - \frac{\sqrt{10}G^{\alpha}_{z}y\left(x-z\right)\left(x+z\right)}{4} - \frac{\sqrt{6}G^{\beta}_{z}y\left(3x^2-2y^2+3z^2\right)}{12}$$

$$\frac{\sqrt{6}G_{xyz}y\left(3x^2-2y^2+3z^2\right)}{12} - \frac{\sqrt{10}G^{\alpha}_{x}z\left(x-y\right)\left(x+y\right)}{4} + \frac{\sqrt{6}G^{\beta}_{x}z\left(3x^2+3y^2-2z^2\right)}{12} + \frac{\sqrt{10}G^{\alpha}_{y}xyz}{2} + \frac{\sqrt{10}G^{\alpha}_{z}x\left(y-z\right)\left(y+z\right)}{4} + \frac{\sqrt{6}G^{\beta}_{z}x\left(2x^2-3y^2-3z^2\right)}{12}$$

$\vec{\mathbb{Q}}^{(3,2)}_{3,1}[g](E,1)$, $\vec{\mathbb{Q}}^{(3,2)}_{3,2}[g](E,1)$

** symmetry

$$x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$$

$$-\frac{3x^2y}{2} + y^3 - \frac{3yz^2}{2}$$

** expression

$$-\frac{7\sqrt{55}G_{xyz}x\left(y-z\right)\left(y+z\right)\left(2x^2-y^2-z^2\right)}{44} + \frac{7\sqrt{55}G^{\beta}_{x}xyz\left(2x^2-y^2-z^2\right)}{22} - \frac{5\sqrt{33}G^{\alpha}_{y}z\left(6x^4-51x^2y^2+5x^2z^2+6y^4+5y^2z^2-z^4\right)}{264} - \frac{\sqrt{55}G^{\beta}_{y}z\left(10x^4+27x^2y^2-29x^2z^2-4y^4-y^2z^2+3z^4\right)}{88} + \frac{5\sqrt{33}G^{\alpha}_{z}y\left(6x^4+5x^2y^2-51x^2z^2-y^4+5y^2z^2+6z^4\right)}{264} - \frac{\sqrt{55}G^{\beta}_{z}y\left(10x^4-29x^2y^2+27x^2z^2+3y^4-y^2z^2-4z^4\right)}{88}$$

$$-\frac{7\sqrt{55}G_{xyz}y\left(x-z\right)\left(x+z\right)\left(x^2-2y^2+z^2\right)}{44} + \frac{5\sqrt{33}G^{\alpha}_{x}z\left(6x^4-51x^2y^2+5x^2z^2+6y^4+5y^2z^2-z^4\right)}{264} + \frac{\sqrt{55}G^{\beta}_{x}z\left(4x^4-27x^2y^2+x^2z^2-10y^4+29y^2z^2-3z^4\right)}{88} - \frac{7\sqrt{55}G^{\beta}_{y}xyz\left(x^2-2y^2+z^2\right)}{22} + \frac{5\sqrt{33}G^{\alpha}_{z}x\left(x^4-5x^2y^2-5x^2z^2-6y^4+51y^2z^2-6z^4\right)}{264} - \frac{\sqrt{55}G^{\beta}_{z}x\left(3x^4-29x^2y^2-x^2z^2+10y^4+27y^2z^2-4z^4\right)}{88}$$

$\vec{\mathbb{Q}}^{(3,2)}_{3,1}[g](E,2)$, $\vec{\mathbb{Q}}^{(3,2)}_{3,2}[g](E,2)$

** symmetry

$$\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$$

$$\frac{\sqrt{15}x^2y}{2} - \frac{\sqrt{15}yz^2}{2}$$

** expression

$$-\frac{\sqrt{33}G_{xyz}x\left(8x^4-40x^2y^2-40x^2z^2+15y^4+30y^2z^2+15z^4\right)}{132} - \frac{7\sqrt{55}G^{\alpha}_{x}xyz\left(2x^2-y^2-z^2\right)}{22} - \frac{\sqrt{55}G^{\alpha}_{y}z\left(4x^4-27x^2y^2+x^2z^2-10y^4+29y^2z^2-3z^4\right)}{88} - \frac{\sqrt{33}G^{\beta}_{y}z\left(60x^4-195x^2y^2-55x^2z^2+60y^4-55y^2z^2+11z^4\right)}{264} - \frac{\sqrt{55}G^{\alpha}_{z}y\left(4x^4+x^2y^2-27x^2z^2-3y^4+29y^2z^2-10z^4\right)}{88} + \frac{\sqrt{33}G^{\beta}_{z}y\left(60x^4-55x^2y^2-195x^2z^2+11y^4-55y^2z^2+60z^4\right)}{264}$$

$$\frac{\sqrt{33}G_{xyz}y\left(15x^4-40x^2y^2+30x^2z^2+8y^4-40y^2z^2+15z^4\right)}{132} - \frac{\sqrt{55}G^{\alpha}_{x}z\left(10x^4+27x^2y^2-29x^2z^2-4y^4-y^2z^2+3z^4\right)}{88} - \frac{\sqrt{33}G^{\beta}_{x}z\left(60x^4-195x^2y^2-55x^2z^2+60y^4-55y^2z^2+11z^4\right)}{264} - \frac{7\sqrt{55}G^{\alpha}_{y}xyz\left(x^2-2y^2+z^2\right)}{22} - \frac{\sqrt{55}G^{\alpha}_{z}x\left(3x^4-x^2y^2-29x^2z^2-4y^4+27y^2z^2+10z^4\right)}{88} + \frac{\sqrt{33}G^{\beta}_{z}x\left(11x^4-55x^2y^2-55x^2z^2+60y^4-195y^2z^2+60z^4\right)}{264}$$

$\vec{\mathbb{G}}^{(3,-3)}_{3,1}[g](E,1)$, $\vec{\mathbb{G}}^{(3,-3)}_{3,2}[g](E,1)$

** symmetry

$$\frac{3x^2y}{2} - y^3 + \frac{3yz^2}{2}$$

$$x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$$

** expression

$$-G_y^\alpha$$

$$G_x^\alpha$$

$\vec{\mathbb{G}}_{3,1}^{(3,-3)}[g](E,2)$, $\vec{\mathbb{G}}_{3,2}^{(3,-3)}[g](E,2)$
** symmetry

$$-\frac{\sqrt{15}x^2y}{2} + \frac{\sqrt{15}yz^2}{2}$$

$$\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$$

** expression

$$G_y^\beta$$

$$G_x^\beta$$

$\vec{\mathbb{G}}_{3,1}^{(3,-1)}[g](E,1)$, $\vec{\mathbb{G}}_{3,2}^{(3,-1)}[g](E,1)$
** symmetry

$$\frac{3x^2y}{2} - y^3 + \frac{3yz^2}{2}$$

$$x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$$

** expression

$$G_{xyz}xz + \frac{\sqrt{15}G_x^\alpha xy}{20} - \frac{G_x^\beta xy}{4} + \frac{\sqrt{15}G_y^\alpha\left(x^2-2y^2+z^2\right)}{15} - \frac{G_y^\beta(x-z)(x+z)}{2} + \frac{\sqrt{15}G_z^\alpha yz}{20} + \frac{G_z^\beta yz}{4}$$

$$-G_{xyz}yz + \frac{\sqrt{15}G_x^\alpha\left(2x^2-y^2-z^2\right)}{15} - \frac{G_x^\beta(y-z)(y+z)}{2} - \frac{\sqrt{15}G_y^\alpha xy}{20} - \frac{G_y^\beta xy}{4} - \frac{\sqrt{15}G_z^\alpha xz}{20} + \frac{G_z^\beta xz}{4}$$

$\vec{\mathbb{G}}_{3,1}^{(3,-1)}[g](E,2)$, $\vec{\mathbb{G}}_{3,2}^{(3,-1)}[g](E,2)$
** symmetry

$$-\frac{\sqrt{15}x^2y}{2} + \frac{\sqrt{15}yz^2}{2}$$

$$\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$$

** expression

$$-\frac{G_x^\alpha xy}{4} - \frac{\sqrt{15}G_x^\beta xy}{4} + \frac{G_y^\alpha(x-z)(x+z)}{2} + \frac{G_z^\alpha yz}{4} - \frac{\sqrt{15}G_z^\beta yz}{4}$$

$$-\frac{G_x^\alpha(y-z)(y+z)}{2} + \frac{G_y^\alpha xy}{4} - \frac{\sqrt{15}G_y^\beta xy}{4} - \frac{G_z^\alpha xz}{4} - \frac{\sqrt{15}G_z^\beta xz}{4}$$

$\vec{\mathbb{G}}_{3,1}^{(3,1)}[g](E,1)$, $\vec{\mathbb{G}}_{3,2}^{(3,1)}[g](E,1)$
** symmetry

$$\frac{3x^2y}{2} - y^3 + \frac{3yz^2}{2}$$

$$x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$$

** expression

$$-\frac{\sqrt{330}G_{xyz}xz\left(x^2-6y^2+z^2\right)}{132}+\frac{5\sqrt{22}G_x^{\alpha}xy\left(x^2+y^2-6z^2\right)}{44}+\frac{\sqrt{330}G_x^{\beta}xy\left(9x^2-5y^2-12z^2\right)}{132}$$
$$-\frac{\sqrt{22}G_y^{\alpha}\left(3x^4-24x^2y^2+6x^2z^2+8y^4-24y^2z^2+3z^4\right)}{88}+\frac{\sqrt{330}G_y^{\beta}\left(x-z\right)\left(x+z\right)\left(x^2-6y^2+z^2\right)}{264}$$
$$-\frac{5\sqrt{22}G_z^{\alpha}yz\left(6x^2-y^2-z^2\right)}{44}+\frac{\sqrt{330}G_z^{\beta}yz\left(12x^2+5y^2-9z^2\right)}{132}$$

$$-\frac{\sqrt{330}G_{xyz}yz\left(6x^2-y^2-z^2\right)}{132}+\frac{\sqrt{22}G_x^{\alpha}\left(8x^4-24x^2y^2-24x^2z^2+3y^4+6y^2z^2+3z^4\right)}{88}-\frac{\sqrt{330}G_x^{\beta}\left(y-z\right)\left(y+z\right)\left(6x^2-y^2-z^2\right)}{264}$$
$$-\frac{5\sqrt{22}G_y^{\alpha}xy\left(x^2+y^2-6z^2\right)}{44}-\frac{\sqrt{330}G_y^{\beta}xy\left(5x^2-9y^2+12z^2\right)}{132}-\frac{5\sqrt{22}G_z^{\alpha}xz\left(x^2-6y^2+z^2\right)}{44}+\frac{\sqrt{330}G_z^{\beta}xz\left(5x^2+12y^2-9z^2\right)}{132}$$

$\vec{\mathbb{G}}_{3,1}^{(3,1)}[g](E,2)$, $\vec{\mathbb{G}}_{3,2}^{(3,1)}[g](E,2)$
** symmetry

$$-\frac{\sqrt{15}x^2y}{2}+\frac{\sqrt{15}yz^2}{2}$$

$$\frac{\sqrt{15}xy^2}{2}-\frac{\sqrt{15}xz^2}{2}$$

** expression

$$-\frac{35\sqrt{22}G_{xyz}xz\left(x-z\right)\left(x+z\right)}{132}-\frac{\sqrt{330}G_x^{\alpha}xy\left(5x^2-9y^2+12z^2\right)}{132}-\frac{5\sqrt{22}G_x^{\beta}xy\left(x^2+y^2-6z^2\right)}{132}-\frac{\sqrt{330}G_y^{\alpha}\left(x-z\right)\left(x+z\right)\left(x^2-6y^2+z^2\right)}{264}$$
$$+\frac{7\sqrt{22}G_y^{\beta}\left(x^4+12x^2y^2-18x^2z^2-4y^4+12y^2z^2+z^4\right)}{264}+\frac{\sqrt{330}G_z^{\alpha}yz\left(12x^2-9y^2+5z^2\right)}{132}+\frac{5\sqrt{22}G_z^{\beta}yz\left(6x^2-y^2-z^2\right)}{132}$$

$$\frac{35\sqrt{22}G_{xyz}yz\left(y-z\right)\left(y+z\right)}{132}-\frac{\sqrt{330}G_x^{\alpha}\left(y-z\right)\left(y+z\right)\left(6x^2-y^2-z^2\right)}{264}-\frac{7\sqrt{22}G_x^{\beta}\left(4x^4-12x^2y^2-12x^2z^2-y^4+18y^2z^2-z^4\right)}{264}$$
$$-\frac{\sqrt{330}G_y^{\alpha}xy\left(9x^2-5y^2-12z^2\right)}{132}-\frac{5\sqrt{22}G_y^{\beta}xy\left(x^2+y^2-6z^2\right)}{132}+\frac{\sqrt{330}G_z^{\alpha}xz\left(9x^2-12y^2-5z^2\right)}{132}-\frac{5\sqrt{22}G_z^{\beta}xz\left(x^2-6y^2+z^2\right)}{132}$$

$\vec{\mathbb{G}}_{3,1}^{(3,3)}[g](E,1)$, $\vec{\mathbb{G}}_{3,2}^{(3,3)}[g](E,1)$
** symmetry

$$\frac{3x^2y}{2}-y^3+\frac{3yz^2}{2}$$

$$x^3-\frac{3xy^2}{2}-\frac{3xz^2}{2}$$

** expression

$$-\frac{7\sqrt{715}G_{xyz}xz\left(x^4-16x^2y^2+2x^2z^2+16y^4-16y^2z^2+z^4\right)}{572}+\frac{35\sqrt{429}G_x^{\alpha}xy\left(2x^4-7x^2y^2+x^2z^2+2y^4+y^2z^2-z^4\right)}{1144}$$
$$-\frac{7\sqrt{715}G_x^{\beta}xy\left(4x^4-19x^2y^2+17x^2z^2+10y^4-43y^2z^2+13z^4\right)}{1144}$$
$$+\frac{5\sqrt{429}G_y^{\alpha}\left(5x^6-90x^4y^2+15x^4z^2+120x^2y^4-180x^2y^2z^2+15x^2z^4-16y^6+120y^4z^2-90y^2z^4+5z^6\right)}{3432}$$
$$+\frac{7\sqrt{715}G_y^{\beta}\left(x-z\right)\left(x+z\right)\left(x^4-16x^2y^2+2x^2z^2+16y^4-16y^2z^2+z^4\right)}{1144}$$
$$-\frac{35\sqrt{429}G_z^{\alpha}yz\left(x^4-x^2y^2-x^2z^2-2y^4+7y^2z^2-2z^4\right)}{1144}+\frac{7\sqrt{715}G_z^{\beta}yz\left(13x^4-43x^2y^2+17x^2z^2+10y^4-19y^2z^2+4z^4\right)}{1144}$$

$$\frac{7\sqrt{715}G_{xyz}yz\left(16x^4-16x^2y^2-16x^2z^2+y^4+2y^2z^2+z^4\right)}{572}$$
$$+\frac{5\sqrt{429}G_x^{\alpha}\left(16x^6-120x^4y^2-120x^4z^2+90x^2y^4+180x^2y^2z^2+90x^2z^4-5y^6-15y^4z^2-15y^2z^4-5z^6\right)}{3432}$$
$$+\frac{7\sqrt{715}G_x^{\beta}\left(y-z\right)\left(y+z\right)\left(16x^4-16x^2y^2-16x^2z^2+y^4+2y^2z^2+z^4\right)}{1144}$$
$$-\frac{35\sqrt{429}G_y^{\alpha}xy\left(2x^4-7x^2y^2+x^2z^2+2y^4+y^2z^2-z^4\right)}{1144}-\frac{7\sqrt{715}G_y^{\beta}xy\left(10x^4-19x^2y^2-43x^2z^2+4y^4+17y^2z^2+13z^4\right)}{1144}$$
$$-\frac{35\sqrt{429}G_z^{\alpha}xz\left(2x^4+x^2y^2-7x^2z^2-y^4+y^2z^2+2z^4\right)}{1144}+\frac{7\sqrt{715}G_z^{\beta}xz\left(10x^4-43x^2y^2-19x^2z^2+13y^4+17y^2z^2+4z^4\right)}{1144}$$

$\vec{\mathbb{G}}_{3,1}^{(3,3)}[g](E,2)$, $\vec{\mathbb{G}}_{3,2}^{(3,3)}[g](E,2)$
** symmetry

$$-\frac{\sqrt{15}x^2y}{2}+\frac{\sqrt{15}yz^2}{2}$$

$$\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$$

** expression

$$\frac{21\sqrt{429}G_{xyz}xz\left(x-z\right)\left(x+z\right)\left(x^2-10y^2+z^2\right)}{572} - \frac{7\sqrt{715}G_x^{\alpha}xy\left(10x^4-19x^2y^2-43x^2z^2+4y^4+17y^2z^2+13z^4\right)}{1144}$$
$$+ \frac{7\sqrt{429}G_x^{\beta}xy\left(4x^4-25x^2y^2+35x^2z^2+4y^4+35y^2z^2-35z^4\right)}{1144} - \frac{7\sqrt{715}G_y^{\alpha}\left(x-z\right)\left(x+z\right)\left(x^4-16x^2y^2+2x^2z^2+16y^4-16y^2z^2+z^4\right)}{1144}$$
$$- \frac{\sqrt{429}G_y^{\beta}\left(13x^6-150x^4y^2-45x^4z^2+60x^2y^4+540x^2y^2z^2-45x^2z^4-8y^6+60y^4z^2-150y^2z^4+13z^6\right)}{1144}$$
$$+ \frac{7\sqrt{715}G_z^{\alpha}yz\left(13x^4+17x^2y^2-43x^2z^2+4y^4-19y^2z^2+10z^4\right)}{1144} - \frac{7\sqrt{429}G_z^{\beta}yz\left(35x^4-35x^2y^2-35x^2z^2-4y^4+25y^2z^2-4z^4\right)}{1144}$$

$$\frac{21\sqrt{429}G_{xyz}yz\left(y-z\right)\left(y+z\right)\left(10x^2-y^2-z^2\right)}{572} + \frac{7\sqrt{715}G_x^{\alpha}\left(y-z\right)\left(y+z\right)\left(16x^4-16x^2y^2-16x^2z^2+y^4+2y^2z^2+z^4\right)}{1144}$$
$$+ \frac{\sqrt{429}G_x^{\beta}\left(8x^6-60x^4y^2-60x^4z^2+150x^2y^4-540x^2y^2z^2+150x^2z^4-13y^6+45y^4z^2+45y^2z^4-13z^6\right)}{1144}$$
$$+ \frac{7\sqrt{715}G_y^{\alpha}xy\left(4x^4-19x^2y^2+17x^2z^2+10y^4-43y^2z^2+13z^4\right)}{1144} + \frac{7\sqrt{429}G_y^{\beta}xy\left(4x^4-25x^2y^2+35x^2z^2+4y^4+35y^2z^2-35z^4\right)}{1144}$$
$$- \frac{7\sqrt{715}G_z^{\alpha}xz\left(4x^4+17x^2y^2-19x^2z^2+13y^4-43y^2z^2+10z^4\right)}{1144} + \frac{7\sqrt{429}G_z^{\beta}xz\left(4x^4+35x^2y^2-25x^2z^2-35y^4+35y^2z^2+4z^4\right)}{1144}$$